Updated Models and Likelihood Construction

TITE-CRM likelihood for fixed-r CRM, random CRM, alpha-CRM, and IPCRM

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1 Purpose

This note documents the likelihood convention for the updated AIDE-CRM models and the event-clock trial simulation used for the recycling strategy. The conclusion for likelihood construction is simple:

All model-based approaches use the TITE-CRM likelihood. For pending non-DLT observations, the contribution is always based on

$$1 - w_i\theta_i,$$

where w_i is the fraction of the assessment window completed and θ_i is the model-implied toxicity probability for row i before TITE weighting. **The delayed-outcome approximation $(1 - \theta_i)^{w_i}$ is not used.**

This applies to:

- fixed-r CRM,
- random-r CRM,
- alpha-CRM,
- IPCRM.

The key pending non-DLT contribution is therefore

$$\boxed{\text{TITE-CRM pending non-DLT term: } 1 - w_i\theta_i.}$$

2 Interim-data notation

At an interim decision time, for patient or patient-cycle row $i = 1, \ldots, N$:

Symbol	Meaning
d_i	Assigned dose level, $d_i \in \{1, \ldots, J\}$.
z_i	Retreat/IPDE indicator. $z_i = 1$ for IPDE/retreat observations and $z_i = 0$ for regular observations.
y_i	Observed DLT indicator by the decision time. Pending patients are coded $y_i = 0$.
u_i	Follow-up time accumulated by the decision time.
T	Full DLT assessment window.

w_i	Fraction of evaluated time, clipped to $[0, 1]$: $w_i = \min\{1, \max(0, u_i/T)\}$. Observed DLTs and fully evaluated patients are forced to $w_i = 1$.
q_j	CRM skeleton probability at dose j.
θ_i	Model-implied marginal toxicity probability for row i before TITE weighting.

3 Common TITE-CRM likelihood

The original TITE-CRM idea uses a weighted dose-response model

$$G(d_i, w_i, \beta) = w_i F(d_i, \beta),$$

so that a partially followed patient contributes through the event probability $w_i F(d_i, \beta)$.

For all updated AIDE-CRM model backends, let θ_i denote the row-specific toxicity probability implied by the selected model before TITE weighting. The TITE-weighted probability is

$$\theta_i^{\text{eff}} = w_i \theta_i.$$

The likelihood contribution is

$$\boxed{L_i = (w_i \theta_i)^{y_i} (1 - w_i \theta_i)^{1-y_i}.}$$

Thus the full likelihood is

$$L = \prod_{i=1}^{N} (w_i \theta_i)^{y_i} (1 - w_i \theta_i)^{1-y_i}.$$

This gives the following cases:

Observation status	Convention	Likelihood contribution
Observed DLT	$y_i = 1,\ w_i = 1$	θ_i
Fully evaluated non-DLT	$y_i = 0,\ w_i = 1$	$1 - \theta_i$
Pending non-DLT	$y_i = 0,\ 0 < w_i < 1$	$1 - w_i \theta_i$
No follow-up yet	$y_i = 0,\ w_i = 0$	1

The delayed-outcome term $(1 - \theta_i)^{w_i}$ is intentionally not used in this updated model document.

4 Fixed-r and random-r CRM: discount model with TITE-CRM likelihood

For the fixed-r and random-r discount CRM backends, the baseline dose-toxicity curve is the power CRM model

$$p_j(\alpha) = q_j^{\exp(\alpha)}, \qquad \alpha \sim \mathcal{N}(0, \sigma_\alpha^2).$$

For a regular observation at dose j, the model-implied probability is

$$\theta_j^{\text{regular}} = p_j(\alpha).$$

For an IPDE/retreat observation at dose j, the carryover-discounted probability is

$$\theta_j^{\mathrm{IPDE}} = r + (1 - r)p_j(\alpha).$$

For row i, define

$$\theta_i = (1 - z_i)p_{d_i}(\alpha) + z_i\theta_{d_i}^{\mathrm{IPDE}}.$$

The TITE-CRM likelihood contribution is

$$L_i(\alpha, r) = (w_i\theta_i)^{y_i}(1 - w_i\theta_i)^{1-y_i}.$$

Therefore,

$$L(\alpha, r) = \prod_{i=1}^{N}(w_i\theta_i)^{y_i}(1 - w_i\theta_i)^{1-y_i}.$$

Fixed-r CRM

For fixed-r CRM, r is user-specified, for example $r = 0.10$. The posterior is proportional to

$$\pi(\alpha \mid D) \propto \left\{\prod_{i=1}^{N}(w_i\theta_i)^{y_i}(1 - w_i\theta_i)^{1-y_i}\right\}\phi(\alpha; 0, \sigma_\alpha^2).$$

Important note. When $r = 0$,

$$\theta_j^{\mathrm{IPDE}} = 0 + (1 - 0)p_j(\alpha) = p_j(\alpha),$$

so regular and IPDE observations have the same model-implied probability. Therefore, fixed-r CRM with $r = 0$ reduces back to the original TITE-CRM likelihood. This is the original TITE-CRM comparator used in the simulations.

Random-r CRM

For random-r CRM,

$$r \sim \mathrm{Beta}(a_r, b_r),$$

and the joint posterior is proportional to

$$\pi(\alpha, r \mid D) \propto \left\{\prod_{i=1}^{N}(w_i\theta_i)^{y_i}(1 - w_i\theta_i)^{1-y_i}\right\}\phi(\alpha; 0, \sigma_\alpha^2)\ \mathrm{Beta}(r; a_r, b_r).$$

JAGS core

The fixed-r and random-r JAGS models use the same likelihood form. The only difference is whether r is supplied as data or sampled from a beta prior.

```
for (j in 1:J) {
  p[j] <- pow(q[j], exp(alpha))
  theta_ipde[j] <- r + (1 - r) * p[j]
```

```
}

for (i in 1:N) {
  theta_base[i] <- (1 - ipde[i]) * p[dose[i]] +
                   ipde[i] * theta_ipde[dose[i]]
  theta_eff[i] <- w[i] * theta_base[i]
  y[i] ~ dbern(theta_eff[i])
}
```

5 Alpha-CRM: cumulative-dose model with TITE-CRM likelihood

The alpha-CRM backend also uses the same TITE-CRM likelihood. The model differs only in how θ_i is constructed.

To avoid notational conflict with the CRM power parameter, denote the deterministic carryover value by $a \in (0, 1)$ and the CRM power parameter by θ.

For patient-cycle row i, define the cumulative effective dose

$$D_i(a) = \sum_{k \le i,\ \text{same patient}} d_k\, a^{(t_i - t_k)/T},$$

where d_k is the actual or scaled dose value from a previous or current treatment row and $t_i - t_k$ is elapsed calendar time.

The function $S\{D_i(a)\}$ is obtained by linearly interpolating the skeleton over the dose-value scale. The model-implied probability is

$$p_i(\theta, a) = S\{D_i(a)\}^{\exp(\theta)}, \qquad \theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2).$$

For alpha-CRM, the row-specific pre-TITE probability is

$$\theta_i = p_i(\theta, a).$$

Therefore the likelihood contribution is exactly the TITE-CRM contribution:

$$\boxed{L_i(\theta, a) = \{w_i p_i(\theta, a)\}^{y_i} \{1 - w_i p_i(\theta, a)\}^{1-y_i}.}$$

Numerically, posterior quantities can be computed by integrating over θ for fixed values of a and then averaging over the numerical integration points. This is only a runtime approximation to the integral. It does not change the modeling convention: the event probability is $w_i p_i(\theta, a)$ and the pending non-DLT term is $1 - w_i p_i(\theta, a)$.

R likelihood core

```
p <- S_at_D^exp(theta)
p_eff <- tite_weight * p
loglik <- sum(y * log(p_eff) + (1 - y) * log1p(-p_eff))
```

This corresponds to

$$\log L(\theta, a) = \sum_i \left[y_i \log\{w_i p_i(\theta, a)\} + (1 - y_i) \log\{1 - w_i p_i(\theta, a)\}\right].$$

6 IPCRM: logistic cumulative model with TITE-CRM likelihood

In the updated helper, IPCRM uses a logistic cumulative model for the row-specific toxicity probability. For row i,

$$\text{logit}(p_i) = \beta_0 + \beta_1 x_i + \beta_2 c_i,$$

where x_i is the assigned dose score and c_i is the cumulative previous-dose score for that patient. The baseline toxicity probability at dose j is

$$\text{logit}(p_j^{\text{base}}) = \beta_0 + \beta_1 s_j,$$

where s_j is the dose score for dose j.

For IPCRM, the row-specific pre-TITE probability is

$$\theta_i = p_i.$$

The updated IPCRM likelihood uses the same TITE-CRM form as the CRM models:

$$\boxed{L_i(\beta_0, \beta_1, \beta_2) = (w_i p_i)^{y_i}(1 - w_i p_i)^{1-y_i}.}$$

Thus a pending non-DLT contributes

$$\boxed{1 - w_i p_i,}$$

not $(1 - p_i)^{w_i}$.

JAGS core for IPCRM

```
for (i in 1:N) {
  logit(p[i]) <- beta0 + beta1 * assigned_d[i] + beta2 * cumu_d[i]
  p_eff[i] <- w[i] * p[i]
  y[i] ~ dbern(p_eff[i])
}
```

The previous delayed-outcome zeros-trick expression

$$y_i \log(p_i) + (1 - y_i) w_i \log(1 - p_i)$$

should not be used in this TITE-only implementation.

7 Comparison of the updated likelihoods

Backend	Pre-TITE model probability θ_i	Pending non-DLT term	DLT term
Fixed-r CRM	$\theta_i = (1 - z_i)p_{d_i} + z_i\{r + (1 - r)p_{d_i}\}$	$1 - w_i\theta_i$	θ_i because $w_i = 1$
Random-r CRM	Same as fixed-r, but $r \sim \text{Beta}(a_r, b_r)$	$1 - w_i\theta_i$	θ_i because $w_i = 1$
Alpha-CRM	$\theta_i = p_i(\theta, a) = S\{D_i(a)\}^{\exp(\theta)}$	$1 - w_i\theta_i$	θ_i because $w_i = 1$
IPCRM	$\theta_i = p_i = \text{expit}(\beta_0 + \beta_1 x_i + \beta_2 c_i)$	$1 - w_i\theta_i = 1 - w_i p_i$	p_i because $w_i = 1$

8 Trial design and recycle strategy

The simulation wrapper uses an event-clock design. Calendar time is tracked explicitly, and each treatment administration row carries its own start time, evaluation time, dose, cycle number, cohort number, and regular-versus-retreat type.

Assignable events and waiting queue

An **assignable event** is an event that can make a treatment administration available for cohort filling. In the current recycle strategy, assignable events are defined as follows:

1. A **new patient arrival** is an assignable event.
2. An **IPDE/retreat decision at the end of a patient's evaluation window** also counts as a new-patient-arrival-type event for the purpose of queueing and cohort filling. If the patient completed the previous evaluation without DLT and remains eligible for another cycle, a retreat event is scheduled at that evaluation time.

Both new-patient events and IPDE-available events are handled through the waiting queue. If an assignable event occurs when no cohort slot is open, the corresponding new patient or IPDE-available patient is added to the waiting queue. When a cohort slot opens, the design fills the slot from this queue rather than ignoring previously arrived patients.

The queue therefore contains two types of pending assignment opportunities:

- `new`: a newly arrived patient who has not yet received treatment;
- `retreat`: a previously treated patient who has completed evaluation and may be available for IPDE/recycling.

The queue is ordered by event time, with a sequence number used to break ties. However, when choosing the next patient for an open cohort slot, the design first checks whether there is an eligible retreat/IPDE event. Eligible retreat patients are assigned before new patients.

Cohort formation and model-based decisions

The design treats patients in cohorts. A cohort is open until `cohortsize` administration rows are assigned. When a cohort is open, the algorithm repeatedly fills available slots from the waiting queue.

Once the cohort is full, the cohort is closed. The design then waits until there is another assignable event. At that time, the current interim dataset is reconstructed using all administrations started by the decision time, their observed DLT status, and their TITE weights. The selected CRM backend is fitted using the common TITE-CRM likelihood, and the next dose is selected subject to the no-skipping and dose-cap rules.

After the model returns the next dose, a new cohort is opened and the waiting queue is again used to fill the available slots. This makes the trial event-driven: decisions are triggered by assignable events and cohort availability, not by fixed calendar intervals.

Recycle and retreat eligibility

The recycling strategy prioritizes eligible retreat/IPDE patients over new arrivals. However, a retreat event is assignable only when all of the following conditions hold:

1. the patient has a previous administration record;
2. the most recent administration was non-DLT;
3. the most recent administration has completed the DLT assessment window;
4. the patient's number of cycles is still below `cycle_max`;
5. the most recent dose was the immediately lower dose, $d_{\text{last}} = d_{\text{current}} - 1$;
6. the current dose is above dose 1.

Therefore, a patient is recycled only after completing the prior cycle without DLT and only when the trial's current dose has moved one level above that patient's most recent dose. This implements the intended AIDE/IPDE strategy: retreat patients are used first when they are eligible, but only for a one-level upward retreat from the immediately lower dose.

If no eligible retreat patient is available, the next new patient in the waiting queue is assigned. If some retreat events are waiting but are not eligible for the current dose, they remain in the queue until a compatible dose decision occurs or until the trial ends.

9 References

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O'Quigley, J., Pepe, M., and Fisher, L. (1990). Continual reassessment method: A practical design for phase I clinical trials in cancer. *Biometrics*, 46, 33–48.